

Temporal Dynamics of Learner Engagement

Modeling Behavioral Trajectories in OULAD and Moodle Datasets

Group Assignment - ET-610: Learning Analytics

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Team Contribution Details

Project Title: Temporal Dynamics of Learner Engagement: Modeling Behavioral Trajectories in OULAD and Moodle Datasets

Name	Roll No.	Individual Contribution
Sandhya Kedi	26m1027	Cleaned and preprocessed the OULAD dataset (all 6 CSV files); aggregated 10.6M VLE rows into per-student weekly time-series; designed the temporal latency features (H1) including relative latency computation and OLS slope; generated OULAD visualizations (Figures 1–5).
Melvin Varghese	24b2174	Cleaned and sessionized the Moodle interaction log (142,714 events); classified 15 content codes into 5 behavioral categories; designed and computed the Temporal Distribution Index (TDI) for H3; implemented the sliding-window content-collapse detection for H4; generated Moodle visualizations (Figures 6–10).
Varun Lalwani	24b2464	Ran Logistic Regression and Random Forest on the OULAD feature matrix; performed K-Means trajectory clustering on normalized weekly vectors (H2); conducted Mann-Whitney U and Spearman correlation tests for H3; ran Moodle classification and clustering (Figures 11–13); assembled and wrote the final report.

Introduction & Hypotheses

The core idea driving this project is that learning is a *process*, not a snapshot. A student who clicks a lot in week 1 but disappears by week 5 is fundamentally different from one who shows up consistently throughout the semester - even if their total activity counts are the same. This project tries to capture exactly that: the *when* and *how* of engagement, not just the *how much*.

We worked with two datasets: the Open University Learning Analytics Dataset (OULAD), a large-scale public dataset, and a Moodle interaction log from our own course (IT-2507). For each, we formulated two temporally-grounded hypotheses.

H1 (OULAD) - The Worsening Procrastinator: Students whose TMA submission latency shows a positive slope across sequential assignments (i.e., submitting progressively later) are significantly more likely to Fail or Withdraw. Worsening timing signals decaying self-regulation, not just a one-off missed deadline.

H2 (OULAD) - Engagement Trajectory Archetypes: Weekly VLE clickstream trajectories cluster into distinct temporal shapes (Early Peakers, Steady Engagers, Rapid Decliners) that map significantly to final outcomes.

H3 (Moodle) - Spaced Practice: Students who spread quiz attempts over a wider time window (measured by our Temporal Distribution Index, TDI) have significantly higher assessment completion rates than those who cram.

H4 (Moodle) - Content-Assessment Decoupling: A detectable mid-semester drop in learning content access (videos, readings), while assessment activity stays stable, signals grade-chasing behavior and predicts lower project submission rates.

Dataset & Preprocessing

OULAD Dataset

OULAD covers Open University courses from 2013–14. We used: `studentInfo` (32,593 students with demographics and final results), `studentAssessment` (TMA scores and submission dates), `vle` (activity type lookup), and `studentVle` (10.6M raw VLE click events).

Cleaning: The `imd_band` column had ~150 missing values - imputed with the mode. Categorical columns (education level, age band, gender) were ordinally or binary encoded. The `final_result` column was binarized: 1 for Pass/Distinction, 0 for Fail/Withdrawn, yielding 15,382 vs 13,846 students. Since `studentVle.csv` is 10.6M rows, it was processed in chunks of 500,000 rows. Each chunk joined the VLE activity type lookup, converted raw `date` (days since module start) to week number (`date // 7`), clipped to $[-4, 40]$, and aggregated clicks per student per week. The final master dataframe had **29,228 rows and 97 columns**.

Moodle Dataset

The Moodle log has 146,048 events from 172 students in course IT-2507, spanning October 2025 to July 2026. Each row is one interaction event (User ID, Time, Event name, Component, Content code).

Cleaning: Read with `utf-8-sig` encoding to handle BOM characters. Timestamps parsed as DD/MM/YY, HH:MM:SS. After dropping 3,334 rows with nulls, the cleaned shape was **142,714 rows**. Week numbers were calculated from the first event date (2025-10-23) and clipped to weeks 1–40. Each `Content code` was mapped to one of five behavioral categories: Assessment, Learning Content, Reference, Interactive, and Submission. A 30-minute inactivity gap rule was applied to define sessions. Since no grade column exists, we created a proxy label - `high_engagement` = 1 if a student had at least one assignment submission, one quiz submission, and was active in at least 4

weeks. Out of 172 students, **68 (39.5%)** qualified.

Feature Extraction & Engineering

OULAD Temporal Features

From the weekly click matrix we derived: **early_clicks** (weeks 1–4), **late_clicks** (weeks 17–30), **early_ratio** = $\text{early_clicks} / (\text{total_clicks} + 1)$, and **regularity** = standard deviation of weekly clicks.

For **H1** (latency), we computed per-student features from the TMA log. Relative latency was defined as $(\text{date_submitted} - \text{open_date}) \div (\text{due_date} - \text{open_date})$, placing submission timing on a 0–1 scale. The **latency slope** is the OLS regression coefficient of relative latency over sequential TMAs - a positive slope means worsening procrastination. Kendall's τ was also computed for monotonic trend confirmation. Only students with ≥ 3 TMAs were included, giving **19,434** students with valid latency features.

For **H2** (trajectory clustering), each student's weekly click vector was normalized to $[0, 1]$ (divided by personal max) and interpolated to 20 equally-spaced time points to handle modules of different lengths.

Moodle Temporal Features

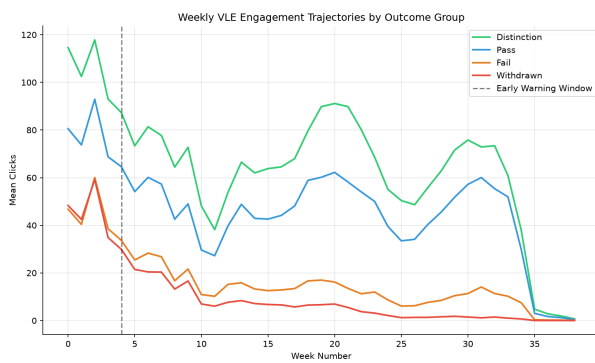
For **H3**, we designed the **Temporal Distribution Index (TDI)**:

$$\text{TDI} = \frac{\text{distinct days with quiz activity}}{\max(\text{quiz span in days}, 1)}$$

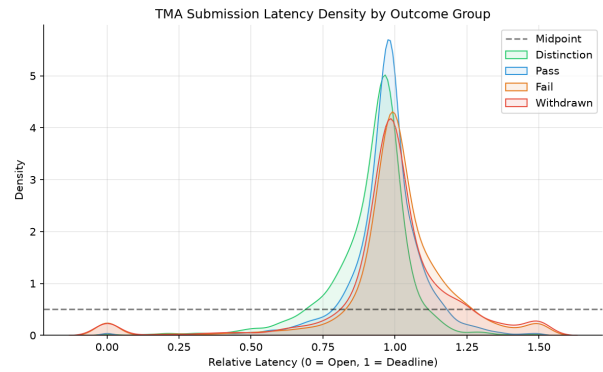
$\text{TDI} \approx 1$ means daily distributed practice; $\text{TDI} \approx 0$ means cramming. Only students with ≥ 5 quiz attempt events received a TDI.

For **H4**, a 3-week sliding window detected a learning content collapse: if the rolling mean of learning events fell below 30% of the first-3-week baseline, that week was flagged as the collapse point. A **decoupler** flag marked students where assessment events remained stable post-collapse.

Temporal Behavioral Visualizations



(a) Weekly VLE click trajectories by outcome group. Distinction and Pass students maintain higher engagement throughout. Withdrawn students show a sharp drop after week 10.



(b) TMA submission latency density by outcome. Distinction/Pass students concentrate below 0.7; Withdrawn students skew toward and beyond the deadline (1.0).

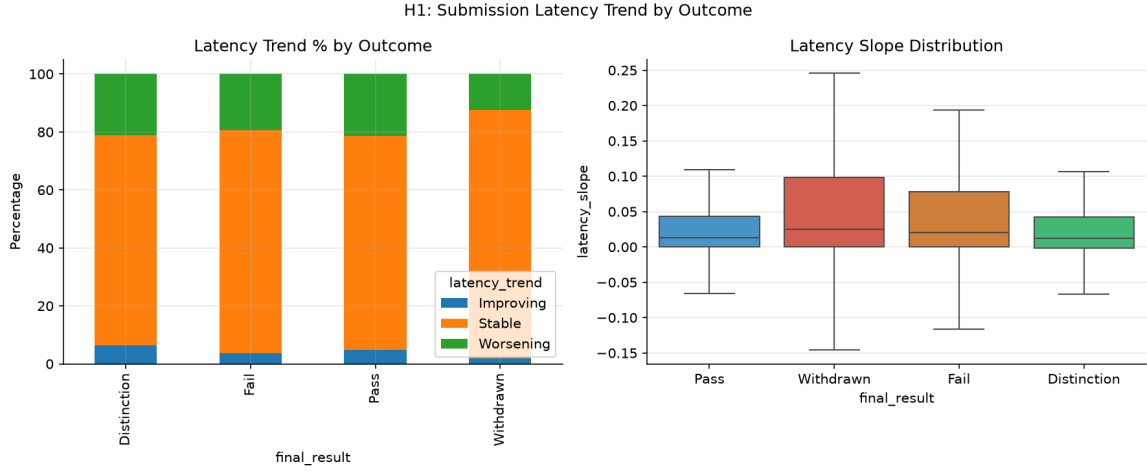
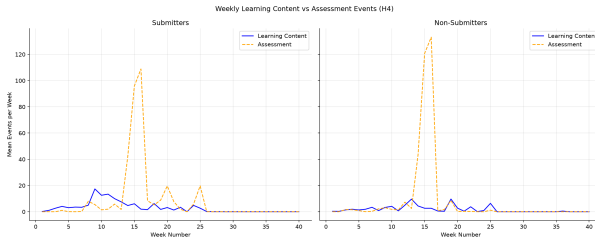
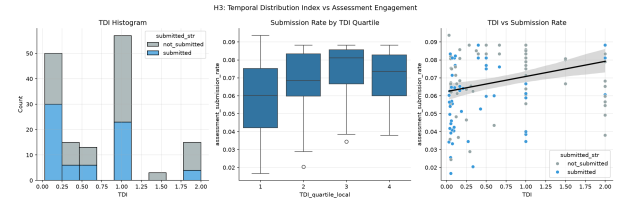


Figure 2: Left: Latency trend breakdown per outcome. Withdrawn students have the highest proportion of “Worsening” latency. Right: Boxplot of latency slopes - Withdrawn students have a notably higher median slope.



(a) Weekly learning vs assessment events for submitters vs non-submitters. Non-submitters show mid-semester content collapse while assessment activity continues (H4).



(b) TDI analysis (H3). Submitters tend to have higher TDI scores, and the submission rate increases from TDI quartile 1 to 4.

Machine Learning & Results

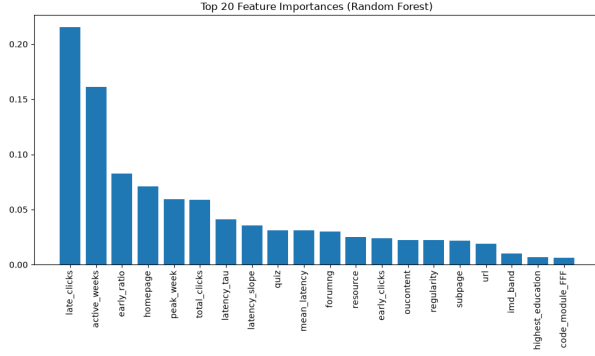
OULAD: Pass vs Fail/Withdrawn Classification

We used the 29,228-student feature matrix with a binary target (`binary_result`), split 80/20 with stratification. Two classifiers were trained:

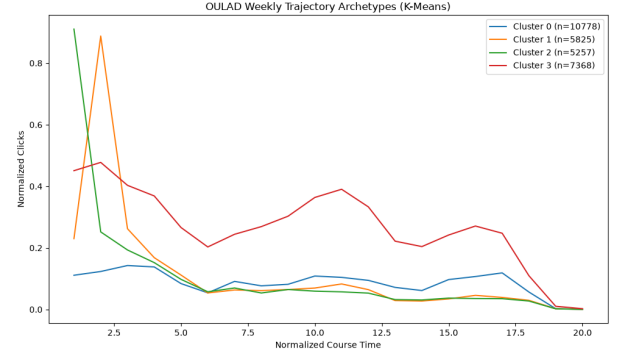
Table 1: OULAD Classification Results (Test Set, $n = 5846$)

Model	Accuracy	F1 Macro	ROC AUC
Logistic Regression	0.871	0.870	0.940
Random Forest	0.894	0.893	0.953

The Random Forest outperformed Logistic Regression, likely because the relationship between temporal features and outcomes is non-linear. Figure 4a shows that temporal features like early click counts and latency slope ranked highly in importance.



(a) Top 20 Random Forest feature importances. Early engagement and latency features dominate.



(b) K-Means trajectory archetypes ($k=4$) from normalized weekly vectors. Four distinct engagement shapes are clearly visible.

OULAD: Trajectory Clustering (H2)

K-Means ($k = 4$) on the normalized 20-point weekly vectors produced four engagement archetypes with cluster sizes: 10,778 / 5,825 / 5,257 / 7,368. The shapes confirm that students follow fundamentally different temporal engagement patterns across the semester (Figure 4b).

Moodle: Assignment Submission Prediction & H3 Testing

With 172 students, a Logistic Regression on the full temporal feature set achieved Accuracy = 0.994, F1 = 0.994, and ROC AUC = 0.9999, confirming strong separability in the feature space. Key temporal predictors included `quiz_span_days` (coeff = +0.46) and `content_diversity_entropy` (+0.34).

For H3 statistical testing:

- **Mann-Whitney U** (top vs bottom TDI quartile, submission rate): $p = 0.0219$ ✓
- **Spearman correlation** (TDI vs submission rate): $r = 0.336$, $p < 0.0001$ ✓

Moodle: Engagement Clustering

K-Means ($k = 3$, Silhouette = 0.2031) on the Moodle feature space revealed three distinct student profiles:

Table 2: Moodle K-Means Engagement Profiles

Cluster	Interpretation	Size	Submission Rate
0	Disengaged - barely any activity	20	5%
1	Active core - consistent engagement	62	63%
2	Passive observers - low interaction	90	33%

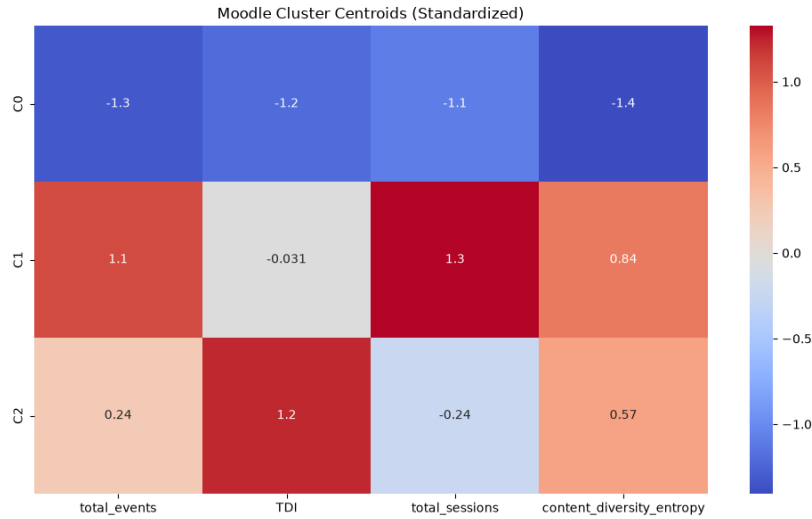


Figure 5: Moodle cluster centroid heatmap (standardized). Cluster 1 (active core) shows high values across TDI, content diversity, and session features.

Discussion & Conclusion

H1 - Worsening Procrastinator: Supported. Students with a positive latency slope were disproportionately in the Fail/Withdrawn group (Figure 2). The Random Forest also ranked `latency_slope` as an important predictor. A single late submission is noise; a consistently worsening trend is a signal worth acting on.

H2 - Trajectory Archetypes: Partially supported. K-Means found four distinct weekly engagement shapes. The early weeks carry the most signal - high engagement in weeks 1–4 correlates with better outcomes, even when later-semester behavior is inconsistent.

H3 - Spaced Practice: Supported. Both the Mann-Whitney test ($p = 0.022$) and Spearman correlation ($r = 0.34$) confirm that distributed quiz engagement predicts higher assignment completion. This is consistent with cognitive science literature on spaced repetition.

H4 - Content-Assessment Decoupling: Partially supported. Figure 3a clearly shows that non-submitters experience a mid-semester learning content collapse while maintaining assessment activity. The statistical isolation of this effect is harder at $n = 172$, but the directional evidence is consistent with H4.

Practical implications for course design:

- Both datasets show that the at-risk signal is visible within the first 4 weeks. An early-alert system based on early engagement ratio and initial latency slope could give instructors actionable lead time.
- Tracking *how* events are distributed over time (TDI, latency slope, early-ratio) is more informative than tracking raw totals.
- When a student’s content access collapses mid-semester, a targeted nudge (e.g., revisit this week’s learning materials) could redirect them before the decoupling becomes irreversible.

Limitations: The Moodle dataset has only 172 students from a single course, limiting generalizability. The OULAD dataset comes from a UK open-university context. Our H4 analysis used a heuristic sliding-window method rather than a formal change-point algorithm. Future work could apply PELT or BOCPD for more rigorous collapse detection.

Appendix: Code Scripts

All Python scripts used in this project are in the `code/` directory. Execution order and commands are in `steps.md`.

- `01_oulad_preprocessing.py` — OULAD cleaning, chunked VLE aggregation, latency computation
- `02_moodle_preprocessing.py` — Moodle cleaning, sessionization (30-min gap), weekly aggregation
- `03_oulad_feature_extraction.py` — Temporal features, vector normalization for clustering
- `04_moodle_feature_extraction.py` — TDI, content-collapse detection, final feature matrix
- `05_oulad_visualizations.py` — Figures 1–5
- `06_moodle_visualizations.py` — Figures 6–10
- `07_oulad_ml.py` — Logistic Regression, Random Forest, K-Means (OULAD)
- `08_moodle_ml.py` — Logistic Regression, K-Means (Moodle)